Food and water born disease

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Definition

- Food borne diseases are as a result of ingestion of food stuffs infected with microorganism or chemicals

- The contamination of food may occur at any stage in the process from production to consumption

Food infections

Contaminated food acting as carrier of microorganisms (Infections include typhoid, cholera, dysentery, hepatitis)

Food that serve as culture medium for growth of pathogen(Infection include salmonellosis, shigellosis,gastroenteristis)

Diseases Transmitted Mainly by Ingestion

1. Bacterial:

Typhoid and paratyphoid fevers
Bacillary dysentery
Gastroenteritis

. Brucellosis
. Anthrax
. Food poisoning

2. Viral:

Poliomyelitis
Diarrhoea diseases

. Viral hepatitis

3. Parasitic;

Protozoa, amebic dysentery and giardia
Helminthes infestation

Diseases Transmitted Mainly by Ingestion

Characters of Water-Borne Epidemic:

Large number of cases - Short time

- Cover localities which share the same source of water
- All age groups - Both sex - All social classes
- When suspected source is removed, epidemic stop
- Epidemic wave of disease proceeded by diarrhoea wave and Followed by secondary wave of cases

Characters of Milk-Borne Epidemic:

- Large number of cases
- Children, old age, pregnant, and lactating women (milk consumers
- High class groups

Remove suspected source, epidemic stop. It is limited to the group of consumers dealt with the suspected food handler

Diseases Transmitted Mainly by Ingestion

Prevention of Diseases Transmitted by Ingestion:

1. Primary level of prevention;

1. General measures:
2. Health education
3. Control of family health
4. Environmental sanitation
5. ***Sanitation requirement of food establishment***

1.1. Specific measures:1. Immunization
2.Chemoprophylaxis

2. Secondary level of prevention:

1. Early diagnosis 2. Prompt treatment

3. Tertiary of prevention, rehabilitation

1. Physical 2. Social 3. Psychological

Diseases Transmitted Mainly by Ingestion

Control of Food handlers:

Person, who prepare, serve, distribute or handle food, he must have a ***valid health license***, to document that he is:- Free from infection by:-

- Clinical examination to exclude, syphilis, tuberculosis, skin infection
- **Stool examination** to exclude typhoid and dysenteries
- **Swab from naso pharynix** to exclude diphtheria
- **Vaccination**
- ***The certificate of license was valid for two years***

Sanitary Requirement of Establishment:

Enteric Fevers

Typhoid fever is a life-threatening illness caused by the bacterium Salmonella Typhi

Reservoir: Man as **a case or carrier**(incubatory, contact, convalescent and healthy carriers).

Susceptibility:

-All ages and severity are highest in younger and adults.

- Both sex, but carrier state is more in female due to increase prevalence of Cholycystitis.

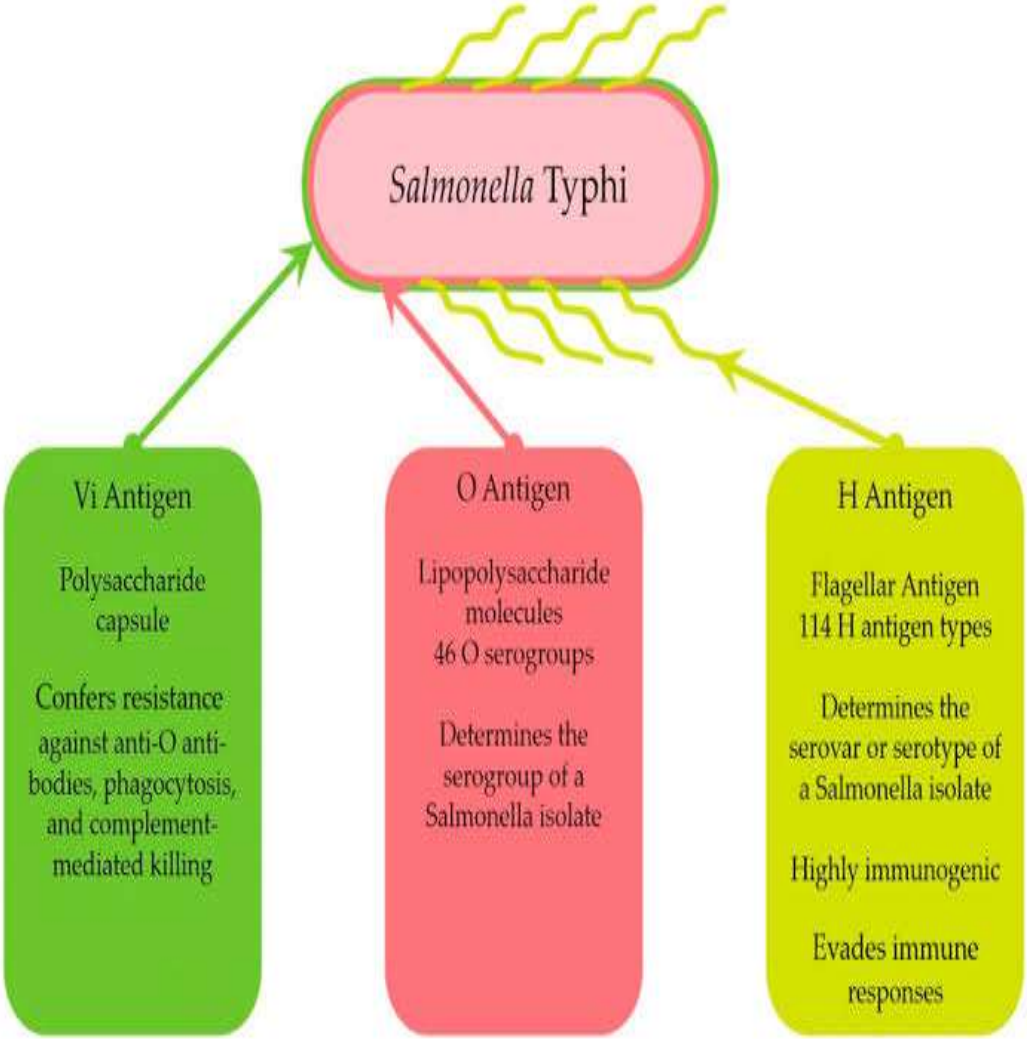
- Seasonal increase in summer

- Social class, more common in rural areas

Aetiology of enteric fever

Salmonella typhi O, H, and Vi antigens

paratyphi A, B and C.



Enteric Fevers

Transmission: ingestion of water, foods, drinks contaminated by human excreta (urine or stool).

Communicability:

People can transmit TF as long as the bacteria remain in their body

most people are infectious prior to and during the first week of convalescence

10% of untreated patients will discharge bacteria for up to 3 months.

Incubation Period: Two weeks, ranges from 1-3 weeks

Enteric Fevers

Clinical Picture:

Acute onset of stepladder fever up to40° C with malaise, anorexia, severe headache, nausea, abdominal pains and constipation or sometimes diarrhoea, rose-coloured spots on the chest due to ruptured capillaries, and enlarged spleen and liver, a typical finding of enteric fever is relative bradycardia for the height of the fever.

Complication: The most frequent complications, which arise with a frequency of 1% to 4% includes gastro-intestinal bleeding and intestinal perforation. Severe neurological forms also have been described with mental dullness, stupor, delirium and shock

- Cholecystitis - Myocarditis - Bronchitis - Meningitis

Fatality: -treated cases is 2 %, untreated cases is 10 %

Enteric Fevers

Diagnosis:

- Clinical, not conclusive.

Laboratory:

- Blood culture: Positive during first week, it is conclusive, Negative does not exclude disease.

2- Widal test: agglutination test, during second week, O antibodies, H antibodies, rising titre is diagnostic. Vi antigen indicating carrier state.

3 - Stool and urine culture; Not conclusive

Prevention:

1. Primary Prevention:

1. General measures:

Health education about personal hygiene, especially regarding hand-washing after toilet use and before food preparation; provision of a safe water supply, sanitary sewage and waste disposal, control of food handlers.

Enteric Fevers

2. Active immunization;

For high risk groups including food handlers, military personnel, camp persons, travellers to endemic areas, contact to cases and occupational workers in sewerage and waste disposal. It no given as routine because it is expensive.

Typhoid vaccines include:

1. TAB vaccine; (The heat-killed vaccine), injectable whole-cell S. typhi vaccine, launched at 1896 - 2 doses 0.5 ml SC then 1 ml SC (4 weeks apart), and 1 ml given every 4 years. Protective effect is 50%.

2. Vi PS vaccine: polysaccharide vaccine, single dose intramascular protective for 2 years, then repeats the dose.

3. Ty21a vaccine: live attenuated vaccine, orally administered (four capsules given every other day over one week, protection for five years.

Protection rate of both vaccines is 65-75%.

Enteric Fevers

2. Secondary Prevention:

1. Cases: - Case finding - Notification - Isolation

-Disinfection, concurrent and terminl - Treatment (Three commonly prescribed antibiotics are ampicillin, trimethoprim-sulfamethoxazole, and ciprofloxacin. - Release, after 3 negative consecutive cultures one day apart

2. Contact: Surveillance for two weeks

- Immunization

3. Epidemic measures: - Sanitary measures as - Super chlorination of water, control of flies, sanitary waste disposal, strict food and milk sanitation.

- Health education - Vaccination of high risk groups or population.

- Epidemiological study to define the source of infection by using bacteriophage typing.